FIT5196 Assessment 1: Exploratory Data Analysis
Group 4

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1. Introduction

This report presents the Exploratory Data Analysis (EDA) conducted on the Group 4 Flickr metadata dataset as part of the FIT5196 Data Wrangling assessment. The dataset comprises 32,892 records of user-uploaded photographs sourced from the Flickr platform, spanning multiple countries and time periods. Each record contains 18 attributes capturing photographic metadata including post identifiers, user identifiers, geospatial coordinates, temporal information, and user-generated textual content such as titles, tags, and descriptions.

The primary objective of this analysis is to systematically examine the structure, quality, and distributional characteristics of the dataset to uncover patterns, inconsistencies, and analytically meaningful relationships across its attributes. The EDA was conducted following a structured pipeline: beginning with dataset validation and missing value profiling, progressing through univariate analysis of numerical, categorical, temporal, and textual attributes, and advancing into bivariate and multivariate analyses to surface relationships between variables.

The findings documented in this report serve a dual purpose. First, they provide transparency into the data wrangling decisions made during Step 1 preprocessing. Second, they establish an empirical foundation for ten key insights and five machine learning research questions that are directly grounded in observed data characteristics.

2. Design of EDA

The EDA was designed as a sequential, layered analytical pipeline applied to the merged and cleaned Group4_dataset.csv file, which was produced during Step 1. The pipeline follows five logical phases, each building on the previous, as described below.

Phase 1 — Structural Validation: The dataset was first inspected for its dimensions, column data types, and the extent of missing values across all 18 attributes. A missing value summary table was produced to quantify null prevalence per column. This phase establishes a baseline understanding of data completeness and correctness before any analytical conclusions are drawn.

Phase 2 — Univariate Analysis: Each attribute was examined independently across four sub-phases corresponding to its data type. Numeric attributes (latitude and longitude) were analyzed using histograms to assess distributional shape and geographic concentration. Categorical attributes (country and city) were examined using frequency bar charts and user activity distributions. Temporal attributes (Taken date, Post date, and derived posting delay) were analysed using time-series bar charts and distribution plots. Text attributes (title, tags, description) were analyzed using character length distributions, tag count distributions, and a top-20 tag frequency chart.

Phase 3 — Bivariate Analysis: Pairs of attributes were analyzed to surface relationships and dependencies. This phase was further divided into three thematic sub-analyses: geographic bivariate analysis (examining how location metadata missingness varies by country), temporal bivariate analysis (examining posting behavior by day-of-week and hour-of-day), and user-level bivariate analysis (comparing tag rate, location rate, and description rate across posts-per-user quantiles).

Phase 4 — Multivariate Analysis: Three compound visualizations were produced to examine interactions between three or more variables simultaneously: a geographic scatter plot colored by metadata availability, a scatter plot of tags count versus posting delay colored by location metadata availability, and a grouped bar chart of metadata completeness rates segmented by user activity type.

Phase 5 — Insight Synthesis: Findings from all four preceding phases were consolidated to derive ten key insights and five machine learning research questions. These were evaluated for their analytical depth, business relevance, and connection to specific EDA observations.

All analysis was implemented in Python using the Pandas, Matplotlib, Seaborn, and NumPy libraries within a Jupyter Notebook environment. Visualizations were produced consistently with labelled axes, titles, and colour schemes appropriate for each chart type.

3. Findings from EDA

3.1 Finding 1: The dataset is substantially incomplete across multiple attributes, with description and tags being the most affected.

```
=== 1.3 MISSING VALUES (NaN string) ===
             NaN_Count  NaN_%
Description      18962   57.6
City             18328   55.7
Country          17541   53.3
Tags             10134   30.8
Title             2178    6.6
Post_ID              1    0.0
User_ID              1    0.0
Secret               1    0.0
Server               1    0.0
Is_Public            1    0.0
Is_Friend            1    0.0
Is_Family            1    0.0
Farm                 1    0.0
Taken_Date           1    0.0
Latitude             1    0.0
Longitude            1    0.0
```

Figure 1. Missing Values Summary Table.

The merged dataset contains 32,892 records across 18 attributes. Inspection of missing values reveals that the three most incomplete attributes are description (57.6% missing, 18,962 records), City (55.7% missing, 18,328 records), and Country (53.3% missing, 17,541 records). The tags attribute is missing in 30.8% of records (10,134), and title is missing in 6.6% (2,178 records). Core identifier fields (Post_ID, User_ID, Secret, Server, Farm) and the binary visibility flags (Is_Public, Is_Friend, Is_Family) each contain exactly one missing record — effectively complete. Notably, latitude and longitude also contain only one missing value each, meaning that coordinate data is near-universally present when a record exists. This stands in contrast to the high missingness in City and Country, which suggests that users frequently upload with GPS coordinates but without completing the named location fields. The heterogeneous missingness pattern confirms that textual and named geographic metadata are user-optional on the Flickr platform, with their absence driven by individual behavior rather than systematic data collection failures.

3.2 Finding 2: The dataset is geographically concentrated in Western Europe, with Sweden and Italy accounting for a disproportionate share of photographs.

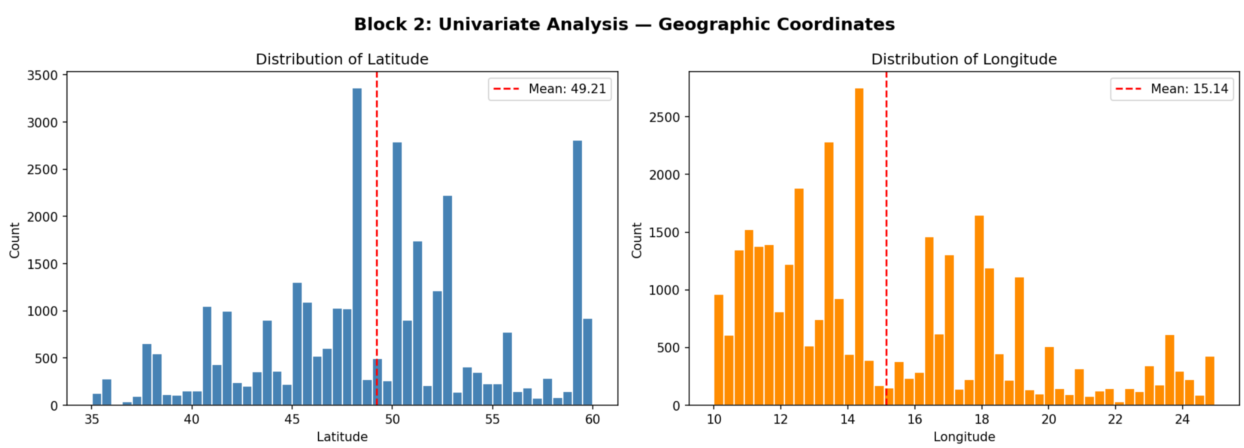
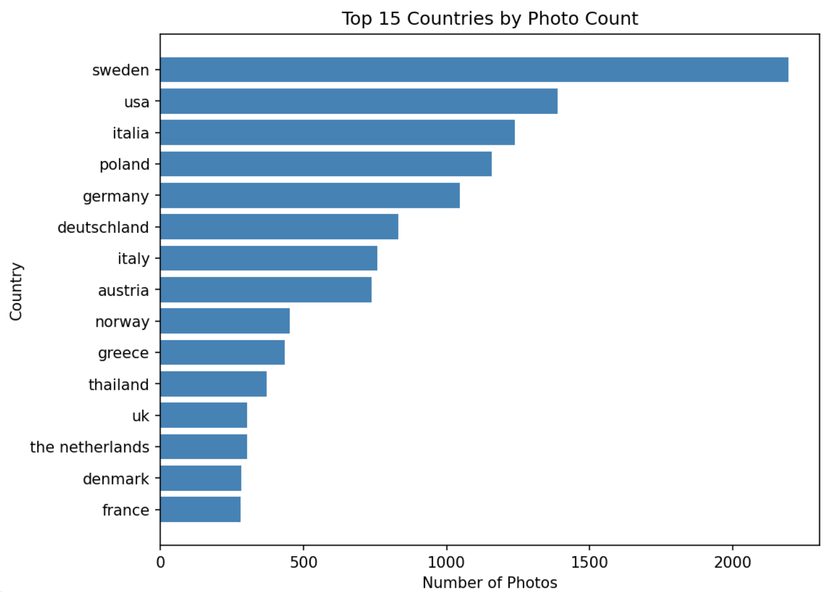


Figure 2. Latitude and Longitude distribution histograms

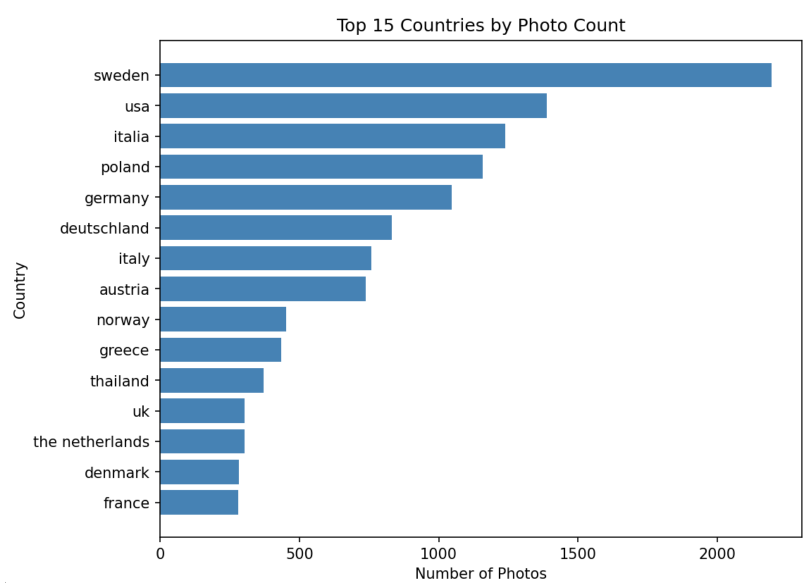



```
=== BLOCK 3: Country Value Counts (Top 15) ===
Country
sweden             2195
usa                1387
italia             1238
poland             1158
germany            1047
deutschland         832
italy               758
austria             739
norway              451
greece              435
thailand            372
uk                  303
the netherlands     302
denmark             282
france              281
```

Figure 3. Top 15 Countries by Photo Count

In Figure 2, all 32,891 geotagged records fall within a latitude range of 35.01°N to 59.99°N (mean 49.21°, std 5.90) and a longitude range of 10.00°E to 24.96°E (mean 15.14°, std 3.77). This coordinate band corresponds precisely to Central and Eastern Europe, encompassing countries such as Sweden, Poland, Germany, Austria, and Italy. Longitude shows moderate positive skew (0.79), indicating a slight concentration toward the lower (western) end of the range. The top 15 country values in Figure 3 confirm this — Sweden leads with 2,195 records, followed by the USA (1,387), Italia (1,238), Poland (1,158), and Germany (1,047). This strong geographic concentration has direct implications for any spatial or cultural modelling task, as the dataset does not represent global Flickr usage.

3.3 Finding 3: Country name inconsistency inflates the apparent diversity of the dataset — Germany and Italy each appear under two distinct language variants.




```
=== BLOCK 3: Country Value Counts (Top 15) ===
Country
sweden             2195
usa                1387
italia             1238
poland             1158
germany            1047
deutschland         832
italy               758
austria             739
norway              451
greece              435
thailand            372
uk                  303
the netherlands     302
denmark             282
france              281
```

Figure 4. Top 15 Countries bar chart (showing 'germany' / 'deutschland' and 'italy' / 'italia') & Counts

The Country attribute contains free-text user-entered values, which has resulted in two major countries being split across language variants. Germany appears as germany (1,047 records) and deutschland (832 records), totalling 1,879 records combined. Italy appears as italia (1,238 records) and italy (758 records), totalling 1,996 records combined. When consolidated, Germany (1,879) and Italy (1,996) both substantially exceed their individual apparent counts, though Sweden (2,195) remains the most represented country in the dataset. The same fragmentation pattern is visible in Figure 4, where germany, deutschland, italy, italia, and itália each appear separately. Any downstream country-level analysis must normalize these variants through entity standardization prior to modelling.

3.4 Finding 4: User posting behavior follows a heavy-tailed power-law distribution, with a small number of highly active users generating the majority of posts.

```
=== BLOCK 3: User Activity ===
Total unique users     : 2346
Max posts by one user  : 1240
Median posts per user  : 2.0
Mean posts per user    : 14.02
```

Figure 5. User Activity Summary Table

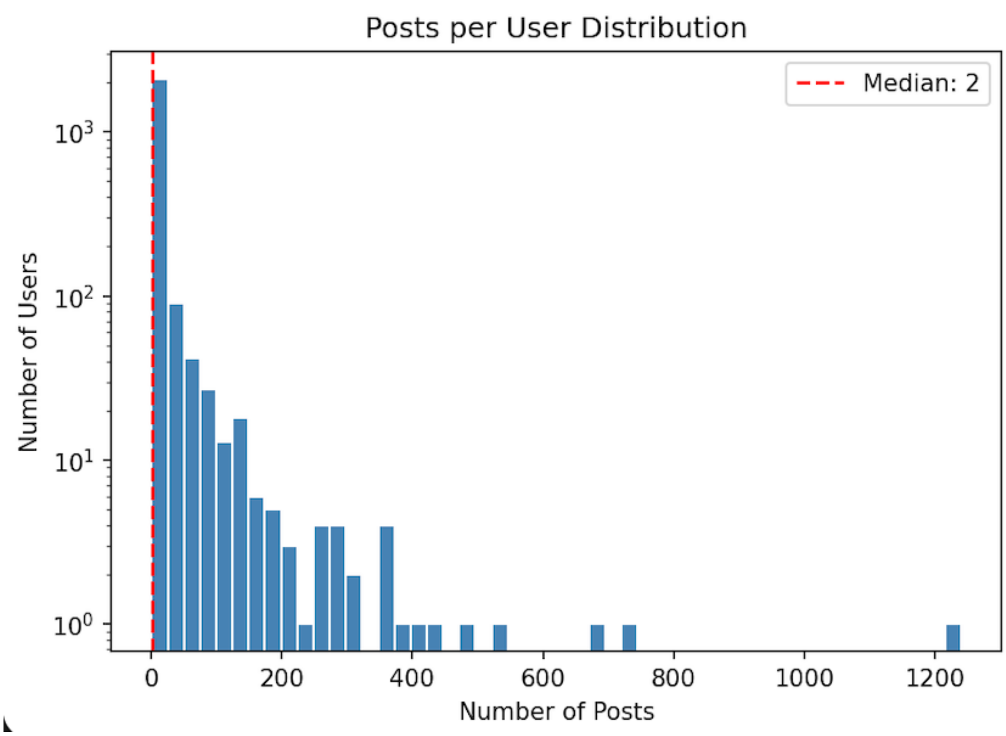
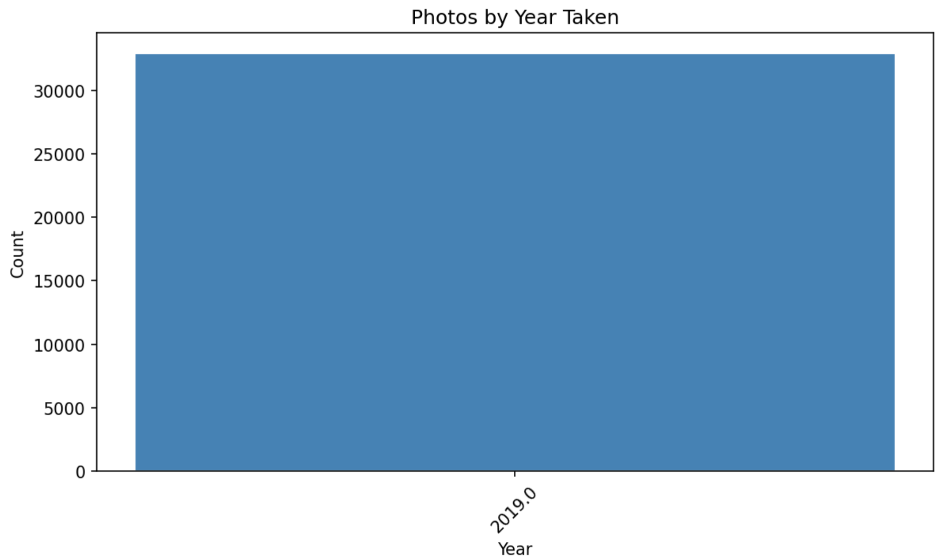


Figure 6. Posts per User Distribution

Figure 6 shows that the distribution of posts per unique user reveals extreme right skew. The median number of posts per user is 2, while the mean is approximately 14.02, and the most active user in the dataset has contributed 1,240 posts (Figure 5). This is a classic power-law (Pareto) distribution commonly observed in user-generated content platforms. Fewer than 5% of users account for the majority of dataset records. This has direct consequences for bias in any user-level analysis — findings driven by the bulk of records may primarily reflect the behavior of a small group of power users rather than the broader user base.

3.5 Finding 5: Photography activity is concentrated in a single two-month window in late 2019, with pronounced weekend and midday peaks.

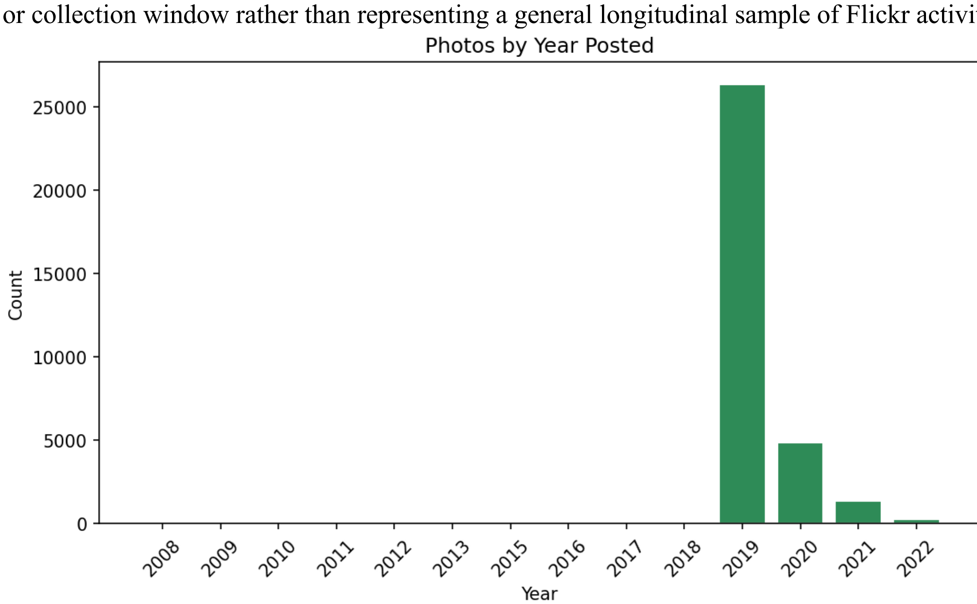



```
Photos by Year Taken:
Taken_Year
2019.0    32891

Photos by Month Taken:
Taken_Month
11.0    28882
12.0     4009
```

Figure 7. Photos by Year Taken bar chart & Count

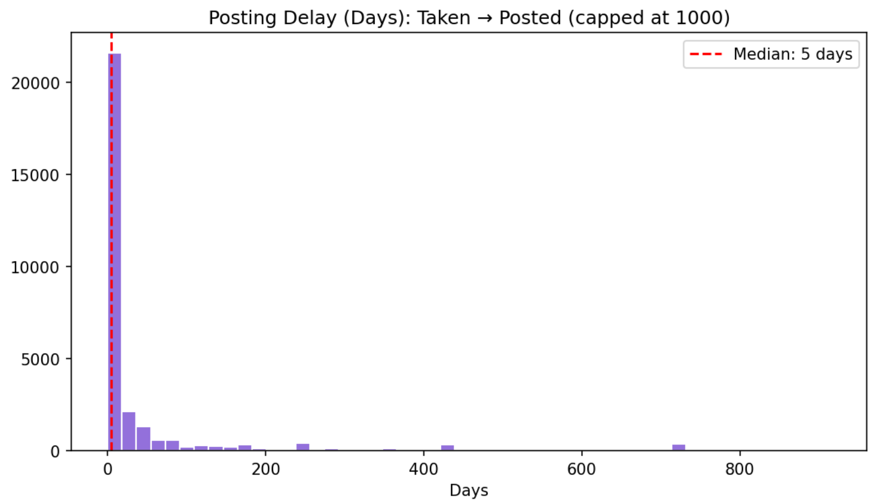
In the above Figure 7, a defining characteristic of this dataset is its extreme temporal concentration in the Taken_Date attribute. All 32,891 records with a valid taken date were photographed in 2019, with 28,882 (87.8%) taken in November and 4,009 (12.2%) in December. This suggests the dataset was assembled from a specific event, project, or collection window rather than representing a general longitudinal sample of Flickr activity.




```
Photos by Year Posted:
Post_Year
2008        4
2009        3
2010       11
2011        2
2012        1
2013        1
2015        2
2016        8
2017        7
2018        6
2019    26348
2020     4885
2021     1345
2022      269
```

Figure 8. Photos by Year Posted Bar Chart

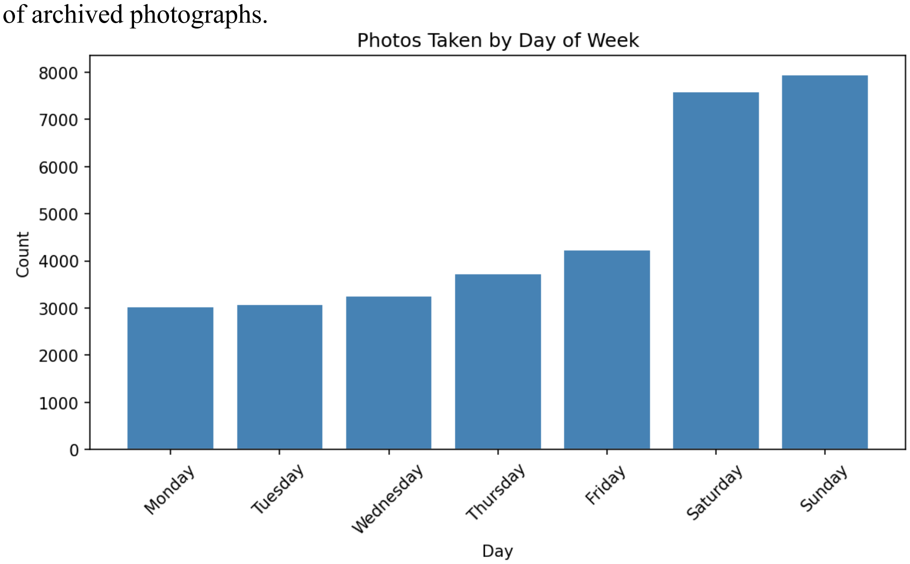
In contrast, Figure 8 shows that Post_Date spans from 2008 to 2022, with the overwhelming majority of posts uploaded in 2019 (26,348 records) and 2020 (4,885 records).




```
Posting Delay Stats (days):
  Min    : 0
  Median : 5
  Mean   : 60.8
  Max    : 915
  % posted same day (0 days): 26.8%
  % posted within 7 days    : 57.5%
  % posted after 1 year     : 5.8%
```

Figure 9. Posting Delay (Days): Taken -> Posted

Figure 9 shows a median delay of 5 days and a mean of 60.8 days, with 26.8% of posts uploaded on the same day the photograph was taken and 57.5% uploaded within 7 days. However, 5.8% of posts were uploaded more than one year after being taken (maximum delay: 915 days), indicating that a subset of users engaged in retrospective bulk uploading of archived photographs.




```
Photos Taken by Day of Week:
Taken_DayOfWeek
Monday       3034
Tuesday      3082
Wednesday    3265
Thursday     3727
Friday       4231
Saturday     7596
Sunday       7956
```

Figure 10. Photos Taken by Days of Week

Figure 10 shows a pronounced weekend bias — Saturday (7,596) and Sunday (7,956) together account for 47.2% of all photographs taken, despite representing only 2 of 7 days.

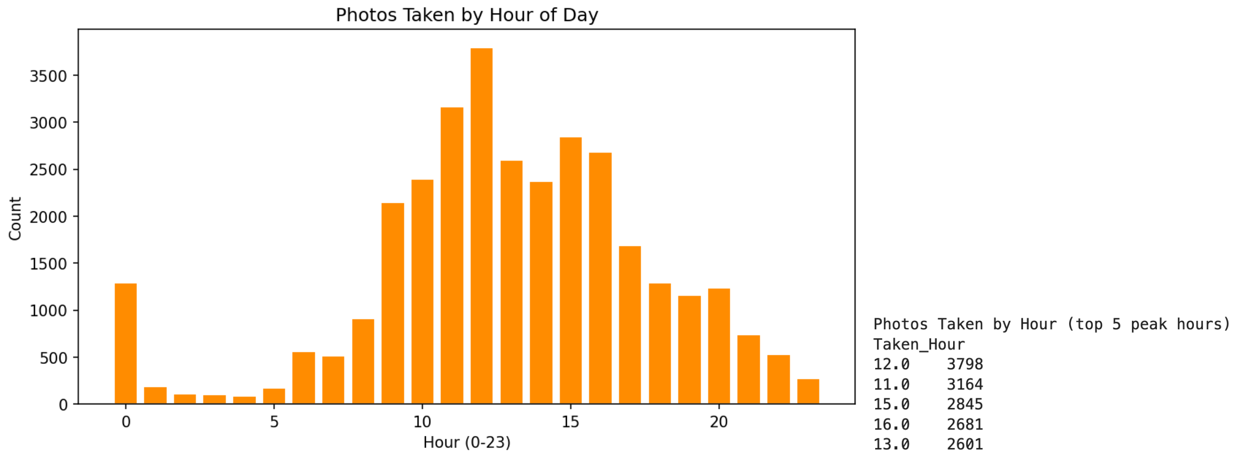


Figure 11. Photos Taken by Hour of Day

Figure 11 shows a midday peak with the highest activity at noon (3,798 photos), followed by 11:00 (3,164) and 15:00 (2,845), consistent with outdoor recreational photography during daylight hours.

3.6 Finding 6: Stockholm is disproportionately the most photographed city, accounting for nearly 2.6 times more posts than the second-ranked city.

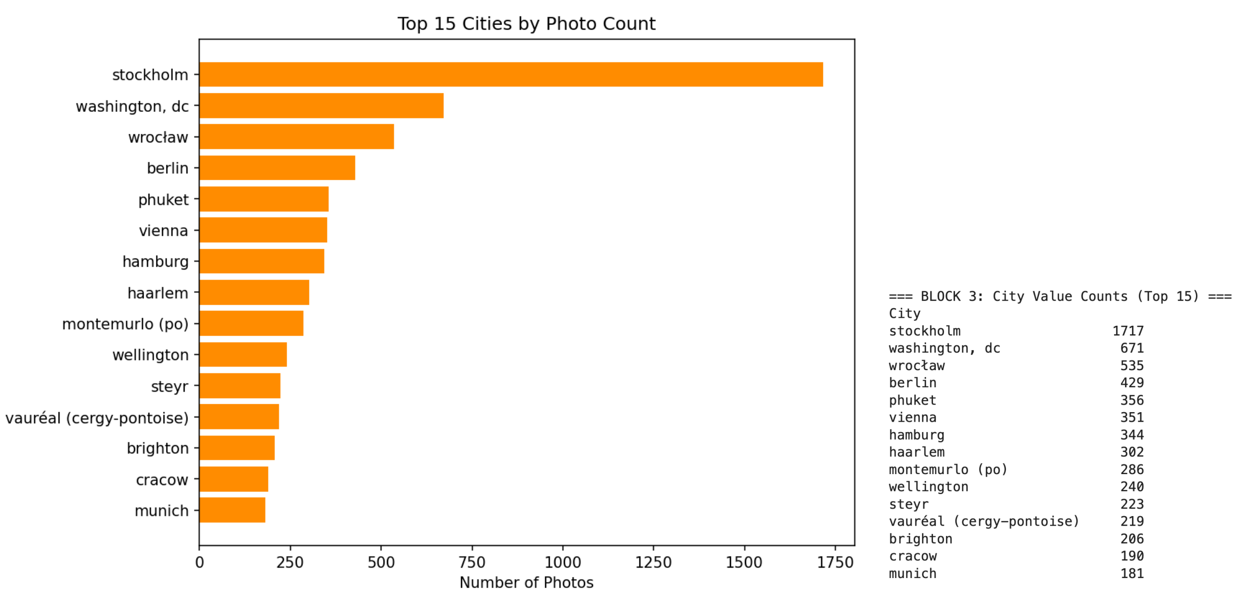
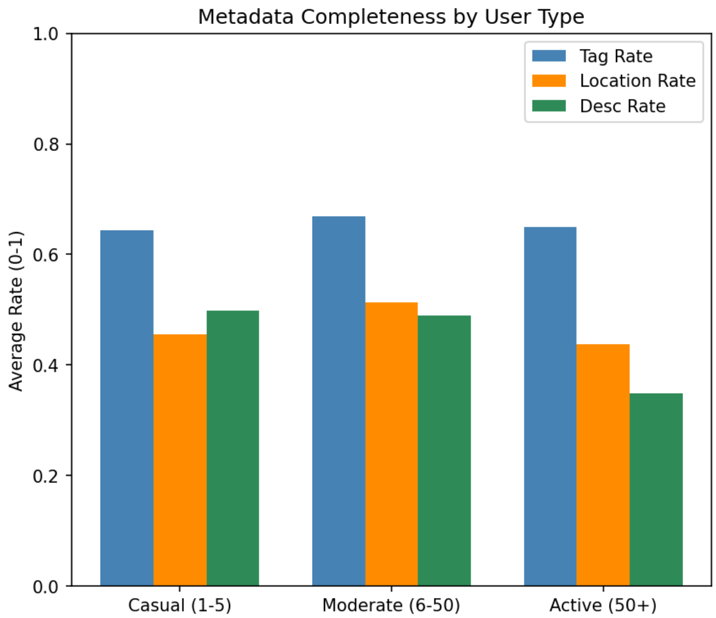


Figure 12. Top 10 Cities Bar Chart & Counts

In Figure 12, among all city values in the City attribute, Stockholm appears 1,717 times — nearly 2.6 times the frequency of Washington DC (671 records), the second most common city. This dominance is notable and not solely explained by Sweden's overall prominence in the dataset. It suggests that Flickr usage was particularly concentrated among Stockholm-based users or active user communities centered in that city during the dataset's collection period. This geographic concentration at the city level warrants consideration in any spatial modelling task.

3.7 Finding 7: More than half of all posts lack a description, and active users — counter-intuitively — provide contextual metadata at a lower rate than casual users.



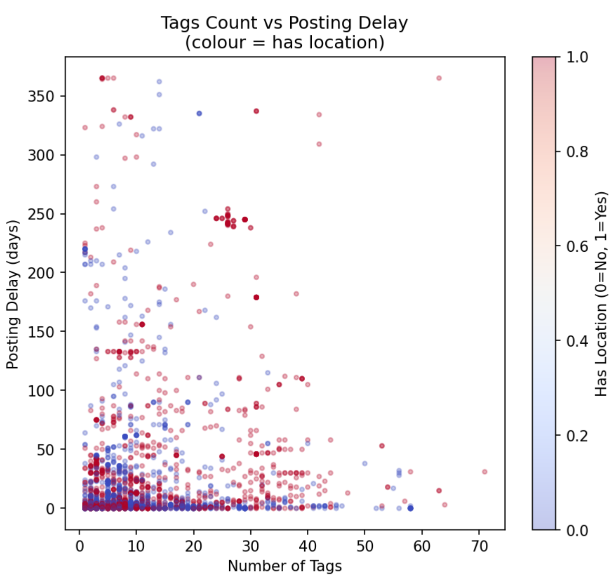

```
=== 1.3 MISSING VALUES (NaN string) ===
             NaN_Count  NaN_%
Description      18962   57.6

=== BLOCK 9: Metadata Completeness by User Type ===
                avg_tags  avg_location  avg_desc
user_type
Casual (1–5)       0.643         0.455     0.498
Moderate (6–50)    0.669         0.514     0.489
Active (50+)       0.650         0.438     0.349
```

Figure 13. Metadata completeness by user type grouped bar chart

Overall, 57.6% of posts contain no description. When broken down by user activity type — Casual (1–5 posts), Moderate (6–50 posts), and Active (50+ posts) — the pattern is counter-intuitive: Casual users maintain a description rate of 0.498 and Moderate users 0.489, while Active users drop significantly to 0.349. Location metadata follows a similar trend: Moderate users have the highest location rate (0.514) compared to Active users (0.438) and Casual users (0.455). Tag usage, by contrast, is relatively stable across all three groups — Casual (0.643), Moderate (0.669), and Active (0.650) — with no meaningful difference. This divergence between tagging behaviour and descriptive/location behaviour suggests that Active users adopt a bulk-upload or archival pattern, prioritising volume over metadata curation. This finding has direct implications for data completeness: the users generating the most records are simultaneously producing the least contextually enriched records.

3.8 Finding 8: Posts with more tags tend to have a longer delay between photograph capture and post upload, though the relationship is weak.




```
=== BLOCK 9: Correlation – Tags Count vs Posting Delay ===
               Tags_Count  Posting_Delay  Has_Location  Has_Tags
Tags_Count          1.000          0.170         0.058       NaN
Posting_Delay       0.170          1.000         0.132       NaN
Has_Location        0.058          0.132         1.000       NaN
Has_Tags              NaN            NaN           NaN       NaN
```

Figure 14. Tags count vs. posting delay scatter plot (colored by location metadata availability)

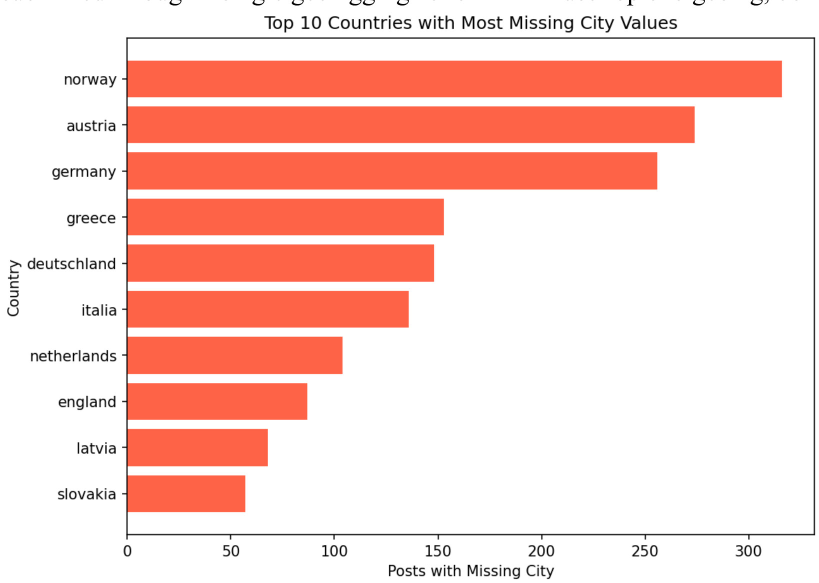
In Figure 14, the Pearson correlation analysis between Tags_Count and posting delay (days between Taken date and Post date) yields a coefficient of 0.170, indicating a weak but statistically meaningful positive relationship. Posts with a high number of tags (30–70) are scattered across posting delays of up to 350 days, while posts with few or no tags tend to cluster near zero delay. A plausible interpretation is that extensively tagged posts require greater manual curation effort, leading users to upload them later. The color dimension (location metadata presence) shows no strong additional pattern, with Pearson correlation of 0.132 between location availability and posting delay — confirming that metadata completeness and posting delay are largely independent.

3.9 Finding 9: Geographic metadata absence is uniformly distributed across European countries and shows no spatial clustering.

```
=== BLOCK 6: City / Country Missingness Crosstab ===
Both present        : 13,146  (40.0%)
Both missing        : 16,123 (49.0%)
City missing only   : 2,205 (6.7%)
Country missing only: 1,418 (4.3%)
```

Figure 15. City/Country missingness crosstab

Figure 15 shows that 49.0% of records (16,123) have both fields missing, while 40.0% (13,146) have both present. Only 6.7% (2,205) have city missing while country is present, and 4.3% (1,418) have country missing while city is present. This near-binary co-occurrence strongly supports the interpretation that location metadata on Flickr is submitted through a single geotagging action — if a user opts to geotag, both fields populate; if not, both are absent.

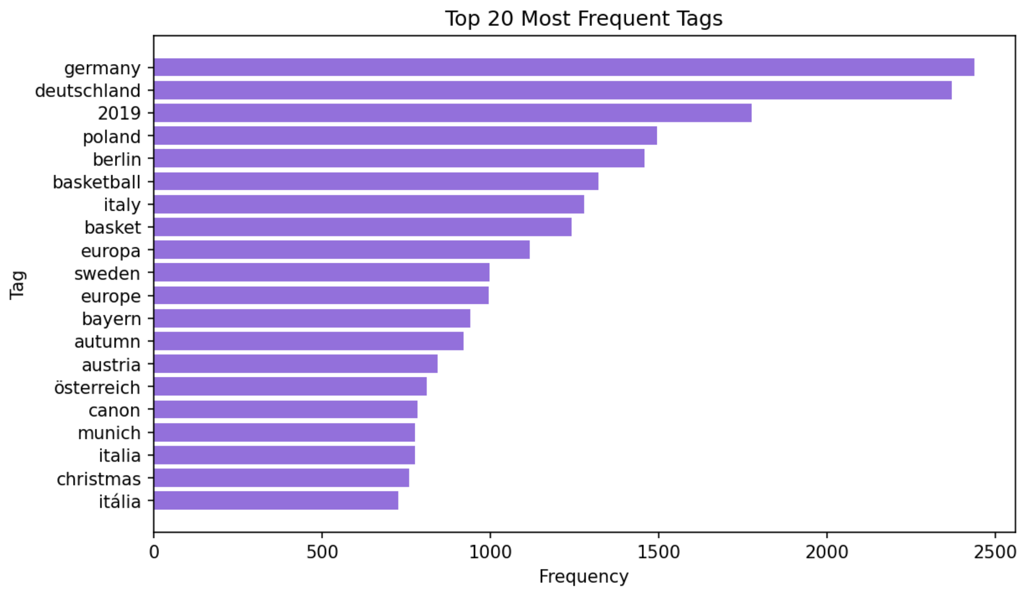



```
=== Top 10 Countries with Most Missing City ===
Country
norway         316
austria        274
germany        256
greece         153
deutschland    148
italia         136
netherlands    104
england         87
latvia          68
slovakia        57
```

Figure 16. Top 10 Countries with Most Missing City Values

Figure 16 shows that amongst the countries with the highest count of records missing city data, Norway (316), Austria (274), and Germany (256) lead, indicating that even records from well-represented countries are not immune to city-level missingness.

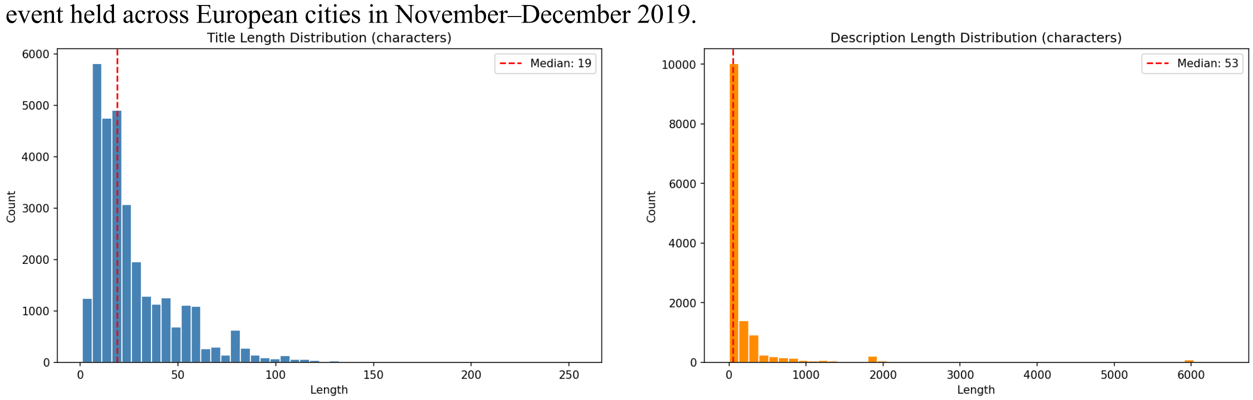
3.10 Finding 10: The top 20 tags reflect subject matter dominated by urban architecture, nature, and events — consistent with the European geographic skew of the dataset.




```
Top 20 Tags:
germany        2437
deutschland    2370
2019           1777
poland         1495
berlin         1459
basketball     1322
italy          1279
basket         1241
europa         1116
sweden          997
europe          994
bayern          940
autumn          921
austria         844
österreich      812
canon           784
munich          776
italia          775
christmas       758
itália          727
```

Figure 17. Top 20 Most Frequent Tags

Figure 17 shows that the top 20 most frequent tags are heavily dominated by geographic place names — germany (2,437), deutschland (2,370), poland (1,495), berlin (1,459), italy (1,279), europa (1,116), sweden (997), europe (994), austria (844), münchen/munich (776/940), and österreich (812). The year 2019 ranks third (1,777 instances), consistent with the temporal finding that all photographs were taken in 2019. Notably, basketball (1,322) and basket (1,241) rank sixth and eighth respectively — a sport-specific cluster that is strikingly prominent relative to other subject-matter tags. The presence of canon (784) suggests a subset of users tag by camera equipment. This tag composition, combining geography, a specific year, and a specific sport, reinforces the hypothesis that the dataset contains a significant event-driven photography component, likely from a basketball tournament or similar sporting event held across European cities in November–December 2019.




```
Title Length:
  Non–NaN count : 30,714
  Mean          : 28.2 chars
  Median        : 19 chars
  Max           : 254 chars

Description Length:
  Non–NaN count : 13,930
  Mean          : 249.6 chars
  Median        : 53 chars
  Max           : 6423 chars
```

Figure 18. Title and Description Length Distributions

Regarding text attribute lengths (Figure 18): titles average 28.2 characters (median 19), descriptions average 249.6 characters (median 53 — highly right-skewed), and posts contain a mean of 11.4 tags (median 8, max 75). The high mean-to-median ratio in description length indicates that a small proportion of users write extensive descriptions while most write brief ones.

4. Key Insights and Machine Learning Research Questions

4.1 Insights

Insight 1: Geographic metadata completeness is a user behavior choice, not a platform limitation. The consistent co-occurrence of missing City and Country fields across all geographic regions confirms that Flickr's location tagging feature was available and functional throughout the dataset's collection period. With 49.0% of records missing both City and Country together and only 11% showing partial mismatch, the data confirms that location metadata is submitted through a single user action — users who opt in provide both fields; those who opt out omit both. The absence of any geographic clustering in missing data rules out infrastructure or API-level explanations.

Insight 2: The effective geographic diversity of the dataset is lower than it appears due to multilingual country naming. With Italy split across italy (758) and italia (1,238), and Germany split across germany (1,047) and deutschland (832), any country-level analysis that does not normalise these variants misrepresents the true distribution. Italy with 1,996 combined records and Germany with 1,879 are both substantially larger than their individual apparent counts, though Sweden (2,195) remains the true leading country in the dataset. The same fragmentation appears in the top-20 tags, making normalization essential before any country-level modelling.

Insight 3: Flickr's user base in this dataset is structured as a long-tail community. The power-law user activity distribution — median of 2 posts per user, mean of 14.0, and a maximum of 1,240 — implies that the dataset's content is disproportionately shaped by a small number of highly active contributors. 35.8% of users have posted only once, while 2.9% of users have posted 100 or more times. Models trained on this data without user-level stratification risk learning the behavior of power users rather than representative Flickr behavior.

Insight 4: High posting volume correlates with reduced metadata richness. Active users (50+ posts) provide descriptions at a rate of only 0.349, compared to 0.498 for Casual users (1–5 posts) and 0.489 for Moderate users (6–50 posts). Location metadata follows the same pattern — Active users have a location rate of 0.438 versus 0.514 for Moderate users. Tag usage remains stable across all three groups (0.643–0.669), indicating that the metadata gap is specifically in descriptive and geographic fields. This creates a systematic relationship between user productivity and data quality that any downstream NLP or geospatial model must account for.

Insight 5: The dataset captures a single concentrated event window in November–December 2019, not a longitudinal sample of Flickr activity. All 32,891 records with a valid taken date were photographed in 2019 — 87.8% in November and 12.2% in December. Combined with basketball (1,322) and basket (1,241) ranking sixth and eighth in the top-20 tags, and the dataset's spread across European cities, this strongly suggests the dataset was compiled from photographs taken at or around a specific pan-European sporting event. This is not a general-purpose Flickr sample. Any model or generalization derived from it must account for the fact that it reflects a specific event context, temporal window, and user community rather than typical platform-wide photography behavior.

Insight 6: Tag count is a weak proxy for user engagement effort but does not reliably predict post timeliness. The weak positive correlation between tag count and posting delay ($r = 0.170$) suggests that higher tagging effort is marginally associated with longer upload delays, but the low coefficient indicates that many other factors dominate — user availability, batch uploading behavior, and content volume. With a mean of 11.4 tags per post and a median of 8, most users apply moderate tagging effort, and the correlation alone is insufficient to predict when a post will go live.

Insight 7: Stockholm's dominance in the cities distribution represents a community concentration effect. Stockholm (1,717 records) appears 2.56 times more frequently than the second-ranked city, Washington DC (671 records), despite Sweden's country-level count being comparable to Italy and Germany. This disproportionate city-level concentration suggests the presence of a large, cohesive user cluster centered in Stockholm rather than a simple consequence of Sweden's overall representation. This community effect is a meaningful signal about how Flickr communities self-organize geographically around specific urban centers.

Insight 8: The absence of descriptions is the largest gap in the dataset's analytical utility. With 57.6% of records (18,962) lacking a description, the description attribute cannot serve as a reliable feature for NLP-based modelling without significant imputation or record exclusion. The 13,930 records that do contain descriptions show high length variance — mean of 249.6 characters but a median of only 53 — indicating that the subset of users who do write descriptions are themselves heterogeneous in their level of engagement. Any text-based model trained on descriptions carries inherent selection bias.

Insight 9: Weekend and midday photography patterns confirm this is primarily a recreational dataset. Saturday (7,596) and Sunday (7,956) together account for 47.2% of all photographs taken despite representing only two of seven days. The hourly peak is at noon (3,798 photos), followed by 11:00 and 15:00 — consistent with outdoor recreational photography during peak daylight hours. This temporal signature is inconsistent with professional or commercial photography workflows. Any business application derived from this dataset should be scoped explicitly to consumer leisure and event-driven photography contexts.

Insight 10: The near-binary co-occurrence of City and Country missingness confirms these fields are populated through a single platform action. Both City and Country are missing together in 49.0% of records (16,123) and present together in 40.0% (13,146), with only 11% of records showing partial mismatch — 6.7% missing City only and 4.3% missing Country only. This near-binary pattern confirms that location metadata on Flickr is submitted as a single geotagging action rather than through independent field-level entry. This has a direct implication for imputation strategy: missingness in one field is a near-perfect predictor of missingness in the other, and any imputation model should treat them as jointly missing rather than independently.

4.2 ML Research Questions

4.2.1 Question 1: Can a machine learning model accurately predict the country of a photograph using geospatial coordinates and textual tag features?

Justification: Finding 2 and Finding 3 established that the dataset is geographically concentrated within Central and Western Europe, yet Country metadata is missing in 53.3% of records — the third most incomplete attribute in the dataset. Since latitude and longitude are present for 32,891 of 32,892 records, and tags frequently contain geographic references such as city and country names (evident from the top-20 tag analysis), a supervised classification model — such as a Random Forest or Gradient Boosted Tree — could leverage both coordinate and tag features to predict missing country values. This question has direct business value: enriching location metadata enables more precise audience targeting, regional content recommendation, and geographic trend analysis on platforms like Flickr. It also contributes to the research domain of geospatial imputation in sparse social media datasets. Success would be measured by macro-averaged F1 above 0.75 on a country-stratified held-out test set, supported by a confusion matrix confirming balanced recall across the top-10 countries — preventing the model from collapsing onto the dominant classes (Sweden, Italy, Germany) at the expense of smaller ones.

4.2.2 Question 2: Can user activity type (Casual, Moderate, Active) be reliably predicted from behavioral and metadata features derived from individual posts?

Justification: Insight 3 and Insight 4 established that user activity type — defined as Casual (1–5 posts), Moderate (6–50 posts), and Active (50+ posts) — is systematically associated with measurable differences in metadata provision behavior. Description rate drops from 0.498 for Casual users to 0.349 for Active users, and location rate follows a similar pattern. A supervised multi-class classification model such as Logistic Regression, SVM, or a decision tree trained on per-post features including tag count, description length, posting delay, and geographic metadata availability could predict user activity tier. This is valuable from a business perspective for platforms like Flickr in segmenting users for personalized engagement strategies — for example, prompting Active users who systematically under-provide descriptions with intelligent metadata suggestions, or identifying Casual users who may benefit from onboarding nudges. Success would be measured by macro-averaged F1 exceeding 0.70, with particular attention to recall on the Active class given its relative rarity (under 3% of users), and ROC-AUC above 0.80 across the three class boundaries to confirm the model is not simply predicting the majority Casual class.

4.2.3 Question 3: Is it possible to predict posting delay — the time between photograph capture and upload — using content and contextual features available at the time of tagging?

Justification: Finding 8 identified a weak but positive correlation ($r = 0.170$) between tag count and posting delay. The dataset shows a median posting delay of 5 days with a mean of 60.8 days, and 5.8% of posts uploaded more than one year after being taken — indicating that upload timing is highly variable even within a dataset concentrated in a single two-month event window (November–December 2019). A regression model such as Ridge Regression or a Gradient Boosted Regressor trained on features including tag count, description length, day and hour of photograph capture, whether location metadata was provided, and user activity type could predict expected posting delay. Given the event-driven nature of this dataset, this question also has relevance to understanding how event photographers manage and release large volumes of content — a behavior with practical implications for content scheduling tools and platform ranking algorithms that weight recency. Success would be defined by a mean absolute error substantially below the naïve baseline of predicting the dataset median (5 days) and an R^2 above 0.30 on a temporally-held-out validation set, recognizing that the high variance in posting delay (up to 915 days) imposes a realistic predictive ceiling well below values typically expected in tightly bounded regression tasks.

4.2.4 Question 4: Can an unsupervised clustering model identify distinct photography communities or thematic groups within the dataset, particularly separating event-driven from general photography?

Justification: Finding 10 identified that the top-20 tags contain a striking sports-specific cluster — basketball (1,322) and basket (1,241) rank sixth and eighth overall — alongside geographic tags and the year 2019. This strongly suggests that the dataset contains at least two distinct photography communities: event photographers documenting a specific sporting event, and general travel or urban photographers. An unsupervised clustering approach such as K-Means or DBSCAN applied to TF-IDF vectors of tag content combined with coordinate and temporal features could identify these latent communities without relying on predefined labels. Separating event photographers from general users has research value in understanding content diversity within photo-sharing platforms and practical value for content recommendation systems that aim to match users with contextually similar communities. Success would be demonstrated by a silhouette coefficient above 0.3 combined with interpretable cluster centroids — specifically, the emergence of at least one cluster dominated by basketball and event-date tags that separates cleanly from clusters dominated by general travel and urban architecture tags, providing qualitative validation that complements the quantitative cohesion score.

4.2.5 Question 5: Can the presence or absence of a photograph description be predicted from user-level and content-level features, and what are the strongest predictors?

Justification: Insight 8 established that description absence is the most analytically limiting gap in this dataset, affecting 57.6% of records (18,962 posts). Rather than treating this purely as a missing data problem, it can be framed as a binary classification task: given features including user activity type, tag count, day and time of upload, country availability, coordinate availability, and title length, can we predict whether a post will have a description? A logistic regression or decision tree classifier would provide an interpretable model revealing the strongest drivers of description provision. Insight 4 already identified user activity type as a meaningful predictor — Active users have a description rate of 0.349 versus 0.498 for Casual users — providing a strong baseline hypothesis for the model. This question contributes to the broader research area of metadata completeness prediction in user-generated content platforms and has practical value for platform designers seeking to reduce metadata sparsity through targeted prompting. Success would be measured by ROC-AUC above 0.75 on a user-stratified held-out test set — stratified at the user rather than post level to prevent information leakage from power users — alongside F1 above 0.65 on the positive class, with feature importance rankings serving as interpretable evidence of the strongest behavioural and content-level drivers of description provision.

5. Conclusion

This report has presented a structured Exploratory Data Analysis of the Group 4 Flickr metadata dataset, comprising 32,892 records and 18 attributes. The analysis was conducted across five systematic phases — structural validation, univariate analysis, bivariate analysis, multivariate analysis, and insight synthesis — producing ten documented findings, ten key insights, and five machine learning research questions.

The EDA revealed several analytically significant characteristics of the dataset. Missing data is pervasive and non-random, concentrated in user-optional fields such as description (57.6%), City (55.7%), Country (53.3%), and tags (30.8%), while core identifiers and geospatial coordinates are near-complete. The dataset is geographically skewed toward Central and Western Europe, particularly Sweden and Italy, with Stockholm exhibiting disproportionate city-level dominance consistent with a concentrated user community. User behaviour follows a heavy-tailed power-law distribution in posting frequency, and high-volume users are systematically associated with lower metadata provision rates — a counter-intuitive finding with direct implications for model training strategies.

Temporal analysis confirmed that this dataset is not a longitudinal Flickr sample but rather a concentrated two-month window in November–December 2019, with pronounced weekend and midday photography peaks indicative of recreational rather than professional use. The prominence of basketball-related tags alongside pan-European geography strongly suggests an event-driven photographic community captured within this window. Textual analysis of tags revealed coherent thematic clusters in subject matter, and the weak but positive correlation between tag count and posting delay ($r = 0.170$) provides a behavioural signal relevant to upload timing prediction.

The five machine learning research questions derived from these findings address geospatial imputation, user activity classification, posting delay regression, unsupervised community discovery, and description presence prediction. Each is grounded in specific, observed data characteristics and carries clear business and research relevance within the Flickr social photography domain. Together, the findings and questions presented in this report constitute a robust foundation for subsequent modelling and analysis tasks.

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